

EC-350 AI and Decision Support Systems

Week 9

Introduction to Machine Learning

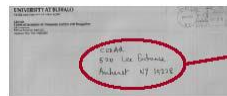
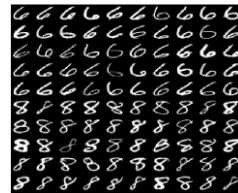
Dr. Arslan Shaukat



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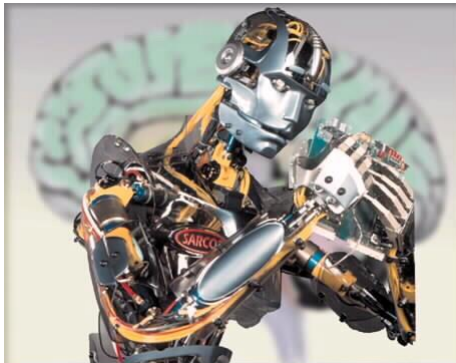
Machine Learning Applications

- Optical Character Recognition (OCR)
 - *Sorting letters by postal code*
 - *Reconstructing text from printed materials*
 - *Handwritten character recognition*
- Analysis and identification of human patterns (Biometric classification)
 - *Face recognition*
 - *Handwriting recognition*
 - *Fingerprints and DNA sequence identification*
 - *Iris scan identification*
 - *Speech recognition/speaker identification*

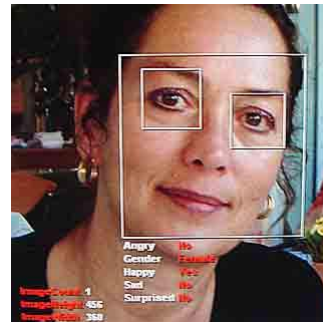


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Applications



amazon.com.



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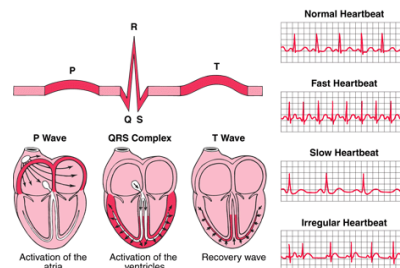
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Examples of ML

- Computer aided diagnosis
 - Medical imaging, EEG, ECG signal analysis
 - Designed to assist (not replace) physicians
- Prediction systems
 - Weather forecasting (based on satellite data)
- Information Retrieval
 - Data Mining



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Machine Perception and Pattern Recognition

- Machine Perception
 - *Build a machine that can recognize patterns*
- Pattern Recognition
 - *Theory, Algorithms, Systems to Put Patterns into Categories*
 - *Relate Perceived Pattern to Previously Perceived Patterns*
- By building such systems, we gain understanding of machine learning, particularly in humans

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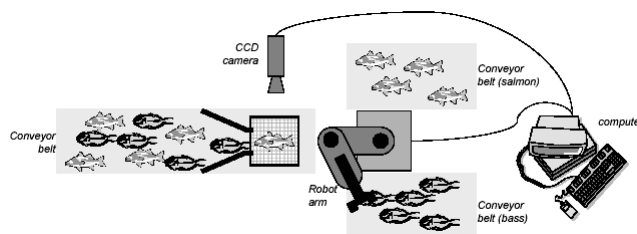
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A Machine Learning Example

- A fish processing plant wants to automate the process of sorting incoming fish according to species (salmon or sea bass)
- The automation system consists of
 - *A conveyor belt for incoming products*
 - *A vision system with an overhead camera*
 - *A computer to analyze images and control the robot arm*



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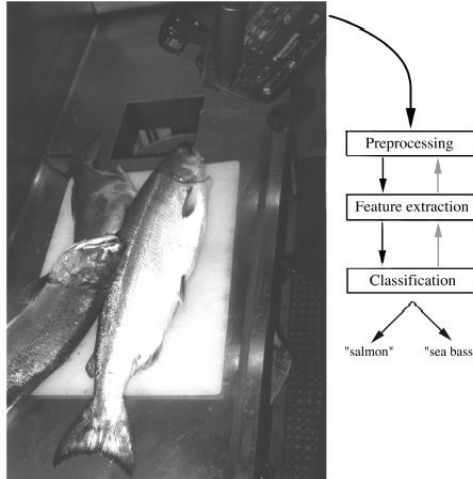
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Example

- “Sorting incoming fish on a conveyor belt according to species using optical sensing”.



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Problem Analysis

- Set up a camera and take some training sample images to extract features.
 - *Length*
 - *Lightness*
 - *Width*
 - *Number and shape of fins*
 - *Position of the mouth, etc...*
- This is the set of all suggested features to explore for use in our classifier
- Purpose:
 - *To classify the future samples based on the data of extracted features from the training samples*

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Example

- Models
 - *There are differences between sea bass and salmon and are viewed as having different models.*
- Preprocessing
 - *Segmentation*
 - *Isolate fish from one another and from the background.*
- Feature Extraction
 - *Information from a single fish is sent to a feature extractor whose purpose is to reduce the data by measuring certain features.*
- Classification
 - *Evaluates the evidence presented and makes a final decision.*

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Selection Criterion

- Suppose sea bass is generally longer than a salmon.
- Select only the length of the fish as a possible feature for discrimination.
- To choose critical value of length, we could obtain some design or training samples of the different types of fish.

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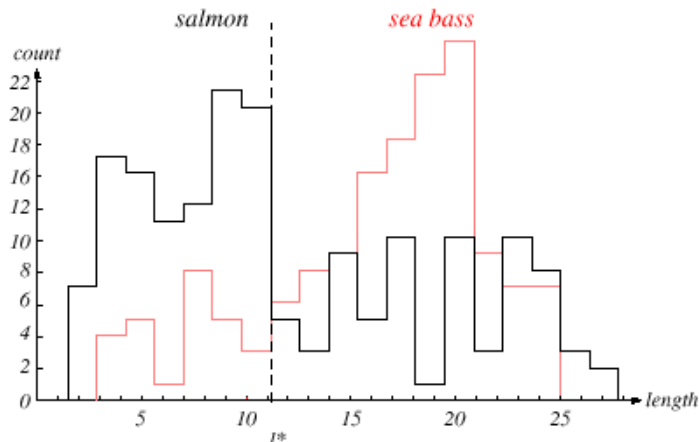
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Histograms for the Length Feature



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Selection Criterion

- No matter how we choose the threshold value of length, we cannot reliably separate sea bass from salmon.
- The length is a poor feature alone!
- Select the average lightness of the fish scales as a possible feature.

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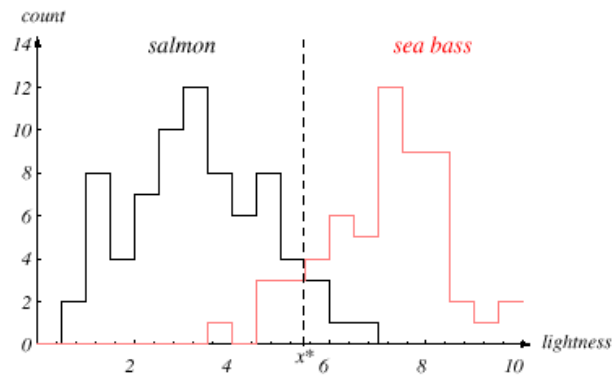
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Histograms for the Lightness Feature



- Length or Lightness, which one is a better feature?
- No value of either feature will “classify” all fish correctly

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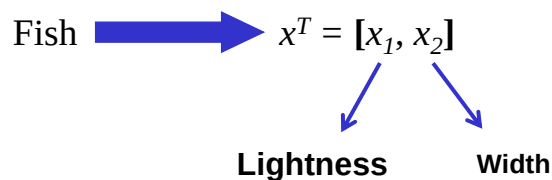
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Selection Criterion and Decision Boundary

- Seek a different feature to separate the fish.
- Use more than 1 feature at a time.
- Adopt the lightness feature.
- Add the width of the fish.
- *Feature vector x is a 2D feature space.*



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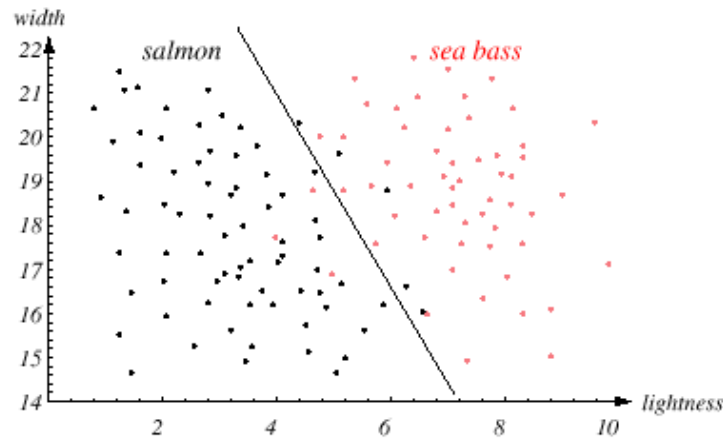
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Two Features of Lightness and Width



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Generalization and Decision Boundary

- We might add other features that are not correlated with the ones we already have, e.g. shape parameters.
- A precaution should be taken not to reduce the performance by adding redundant features.
- Ideally, the best decision boundary should be the one which provides an optimal performance such as in the following figure:

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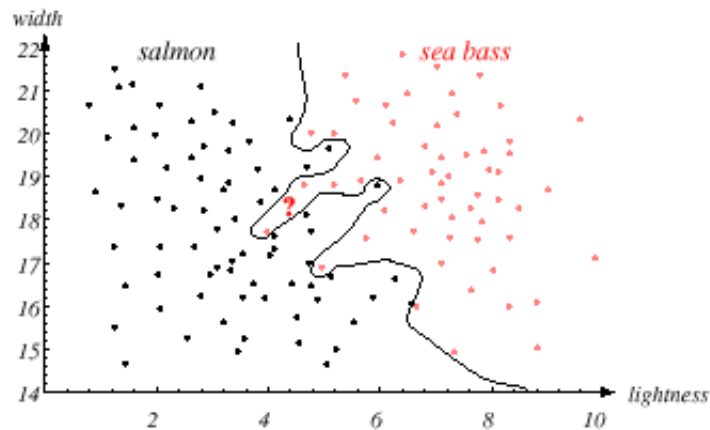
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Complex Decision Boundary



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Generalization and Decision Boundary

- However, our satisfaction is premature because the central aim of designing a classifier is to correctly classify novel input



Generalization!

- It is unlikely that the complex decision boundary would provide good generalization.

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Generalization and Decision Boundary

- More training samples for estimating the true characteristics of the categories.
- Amount of data in most problems is limited.
- Even with vast data, the classifier can give a complicated decision boundary.
- A simple classifier with non-complex decision boundary can provide good generalization.

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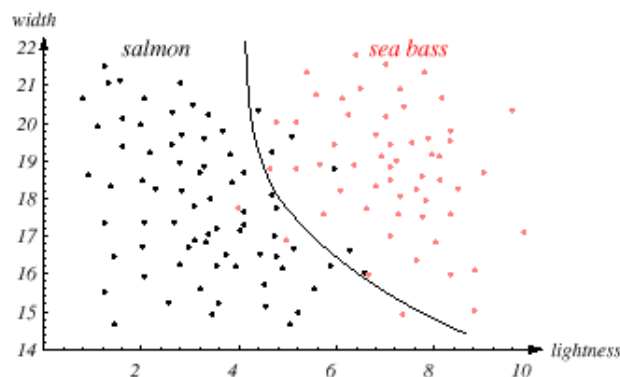
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Selected decision boundary

- The decision boundary can be a simple curve which might represent the optimal trade-off.



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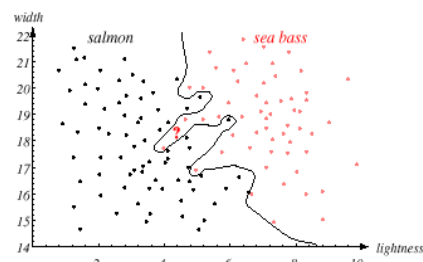
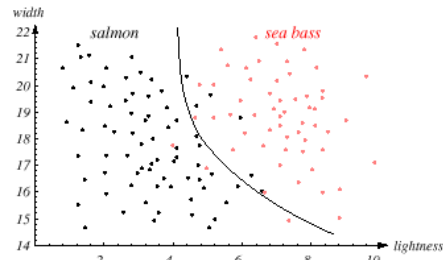
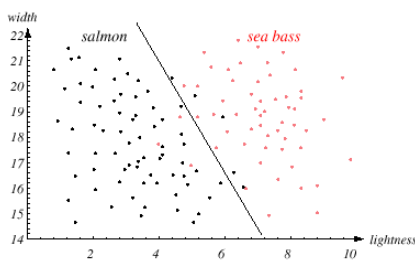
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Decision Boundary Selection



- Which of the boundaries would you choose?
 - Simple linear boundary – training error > 0
 - Nonlinear complex boundary – training error = 0
 - Simpler nonlinear boundary – training error > 0

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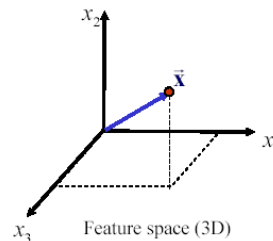
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Terminologies in Machine Learning

- **Features:** a set of variables believed to carry discriminating and characterizing information about the objects under consideration
- **Feature vector:** A collection of d features, ordered in some meaningful way into a d -dimensional column vector, that represents the signature of the object to be identified.
- **Feature space:** The d -dimensional space in which the feature vectors lie. A d -dimensional vector in a d -dimensional space constitutes a point in that space.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \begin{matrix} \text{feature 1} \\ \text{feature 2} \\ \\ \text{feature } d \end{matrix}$$



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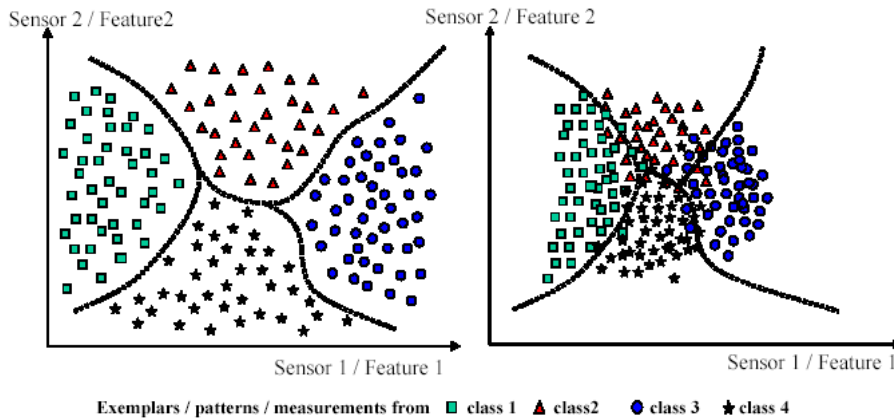
Terminologies in ML

- **Class:** The category to which a given object belongs
- **Decision boundary:** A boundary in the d -dimensional feature space that separates patterns of different classes from each other
- **Training Data:** Data used during training of a classifier for which the correct labels are *a priori* known
- **Testing Data:** Unknown data to be classified. The correct class of this data are not known *a priori*

Terminologies in ML

- **Classifier:** An algorithm which adjusts its parameters to find the correct decision boundaries –through a learning algorithm using a training dataset
- **Error:** Incorrect labelling of the data by the classifier
- **Training Performance:** The ability/performance of the classifier in correctly identifying the classes of the training data, which it has already seen. It may not be a good indicator of the generalization performance.
- **Generalization (Test Performance):** The ability/performance of the classifier in identifying the classes of previously unseen

Kinds of Data



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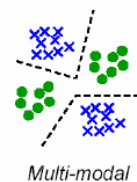
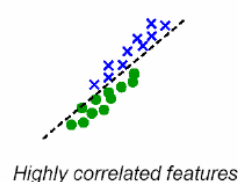
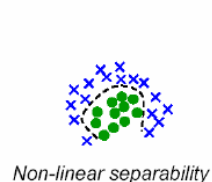
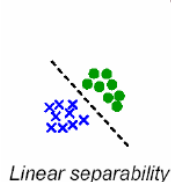
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Good Features vs. Bad Features

- Ideally, for a given group of patterns coming from the same class, feature values should all be similar
- For patterns coming from different classes, the feature values should be different



More feature properties



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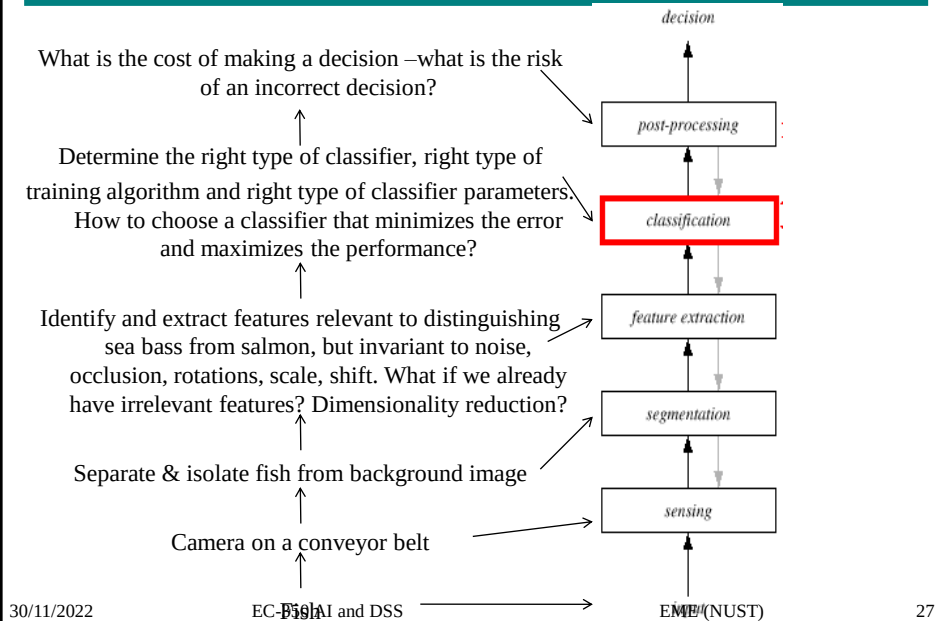
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Components of Machine Learning System



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Machine Learning Systems

- Sensing
 - Use of a transducer (camera or microphone).
 - ML system depends on the bandwidth, resolution, sensitivity, distortion, etc. of the transducer.
- Segmentation
 - Patterns should be well separated and should not overlap.
- Feature extraction
 - Distinguishing features
 - Invariant features with respect to translation, rotation and scale.

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Machine Learning Systems

- Classification
 - *Use a feature vector provided by a feature extractor to assign the object to a category.*
 - *Not always possible to determine the values of all the features.*
- Post Processing
 - *Post-processor uses the output of the classifier to decide on the recommended action.*
 - *Error rate*

Learning and Adaptation

- Learning incorporates information from training samples in classifier design.
- It refers to some form of algorithm for reducing the error on training data.
- Supervised learning
 - *A teacher provides a category label for each pattern in the training set.*

Learning and Adaptation

- Unsupervised learning
 - *The system forms clusters or “natural groupings” of the input patterns.*
 - *The labels of the categories are unknown.*
- Reinforcement Learning
 - *Learning with a critic.*
 - *No desired category signal is given; instead the only teaching feedback is that the tentative category is right or wrong.*

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Conclusion

- Overwhelmed by the number, complexity and magnitude of the sub-problems of Machine Learning.
- Many of these sub-problems can indeed be solved.
- Mathematical theories solving some of these problems have in fact been discovered.
- Many fascinating unsolved problems still remain.

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